

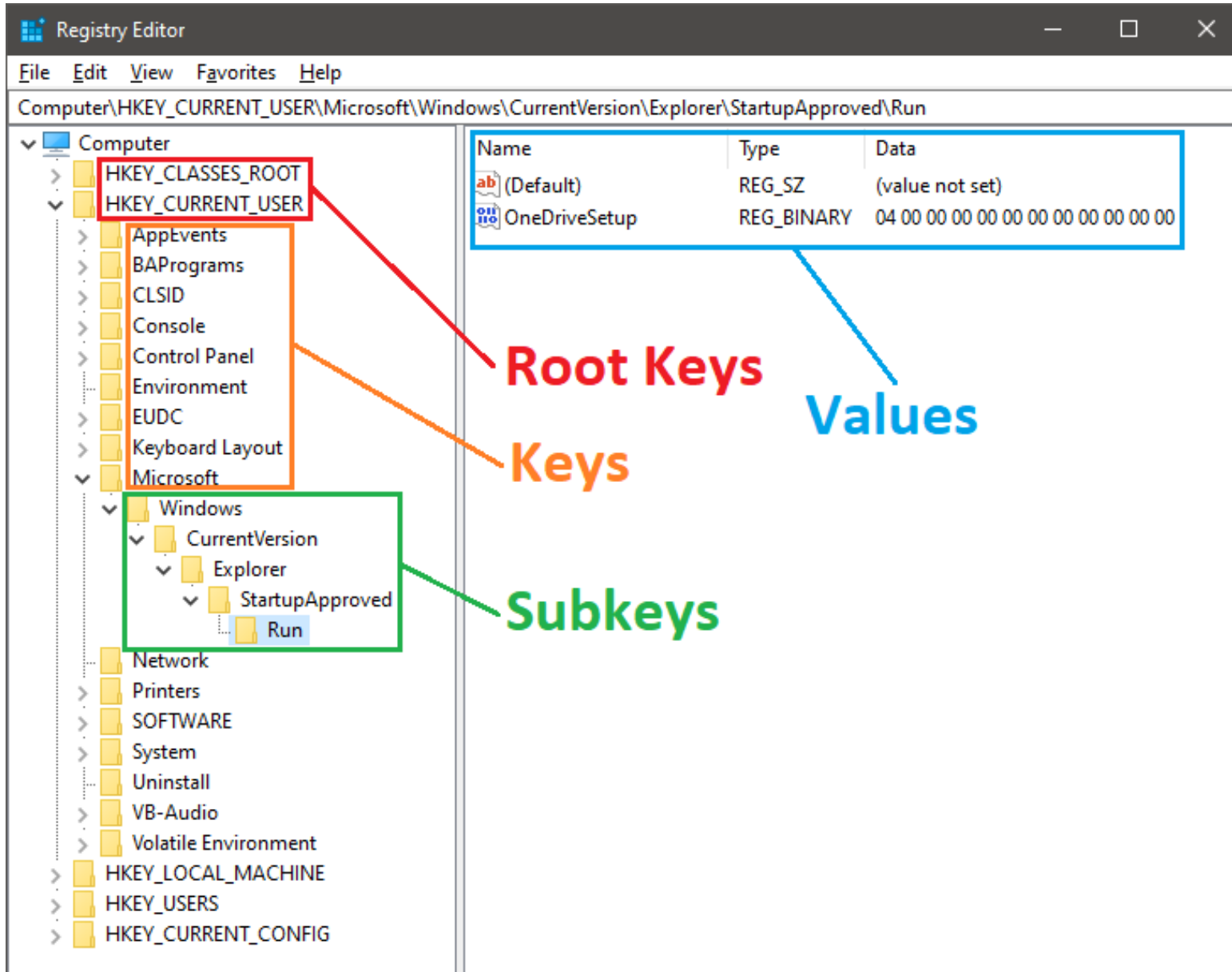
MODULE 04

WINDOWS & LINUX FORENSICS

Windows Registry Structure

- The Windows Registry is a hierarchical database that stores configuration settings and options for the operating system, installed applications, and user preferences.
- It is divided into major root keys such as HKEY_LOCAL_MACHINE (HKLM), which contains system-wide settings, and HKEY_CURRENT_USER (HKCU), which stores user-specific configurations. Each root key contains subkeys, similar to folders, and values, which store the actual data.
- Forensic investigators rely heavily on the registry because it provides persistent evidence of system activity, including installed programs, connected devices, recently opened files, and user logins.
- Even if a user deletes files, traces often remain in registry keys.

Windows Registry Structure



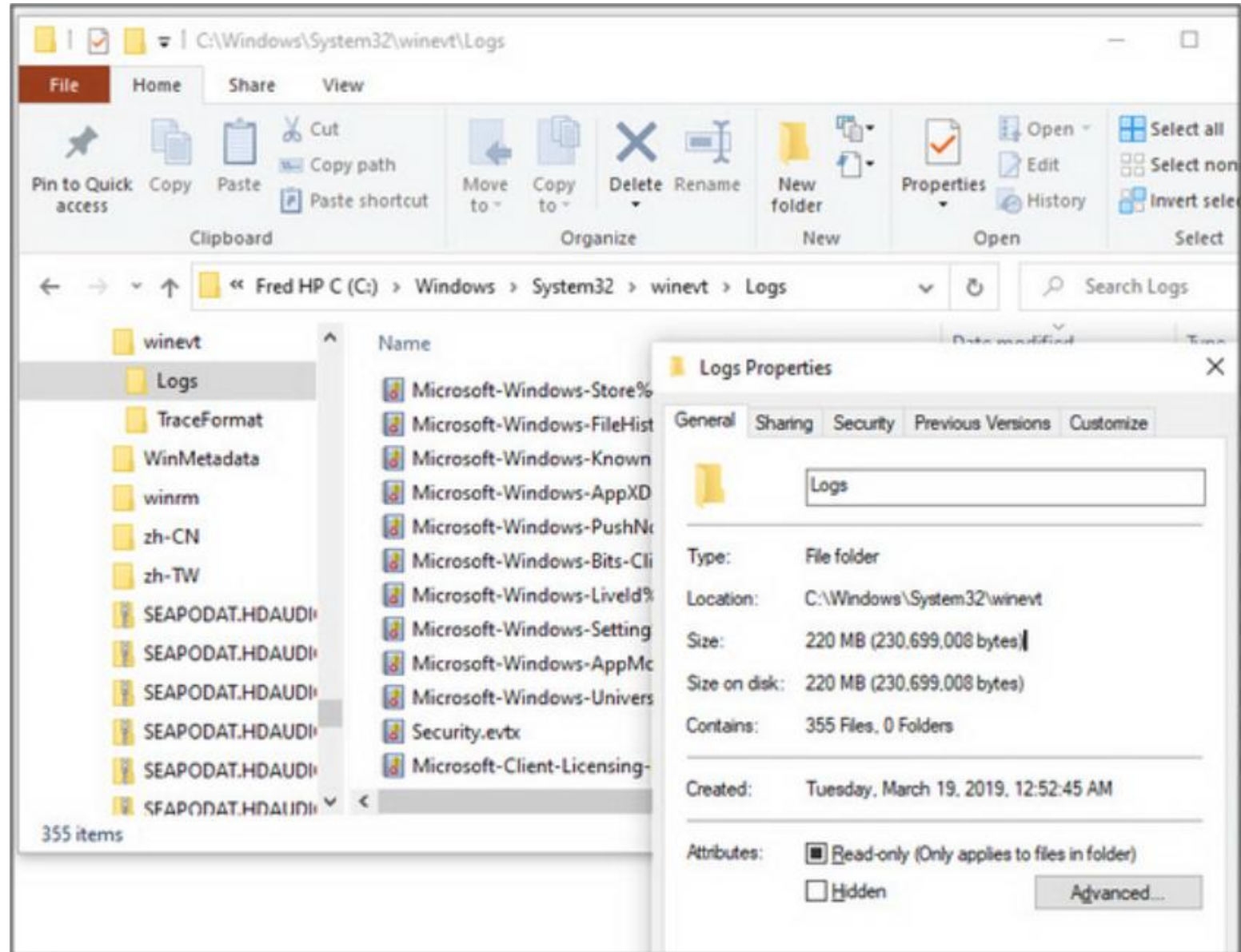
Windows Registry Structure

- From a forensic standpoint, specific registry paths are especially valuable.
- For example, the “Run” keys can reveal programs configured to execute at startup, while the “USBSTOR” key logs USB device connections.
- Investigators can reconstruct timelines by analyzing last modified timestamps of registry keys.
- Since the registry is stored in hive files (e.g., SYSTEM, SOFTWARE, SAM, NTUSER.DAT), these can be extracted and analyzed offline using forensic tools.

Windows Event Logs

- Windows Event Logs are records maintained by the operating system that document system, security, and application events. These logs are stored in **.evtx** format and can be accessed through the Event Viewer.
- The three primary categories are **System logs** (hardware and OS events), **Application logs** (software-related events), and **Security logs** (authentication and authorization activities).
- Security logs are particularly crucial in forensic investigations because they capture login attempts, privilege use, and account changes.
- Each log entry includes detailed metadata such as **event ID, timestamp, source, and user account**. Investigators use event IDs to identify specific activities—for example, Event ID 4624 indicates a successful login, while 4625 signals a failed login attempt.
- Correlating multiple log entries allows analysts to reconstruct sequences of events, detect anomalies, and identify malicious behavior such as brute-force attacks or unauthorized access.

Windows Event Logs



QnA

- In a corporate data theft investigation, analysts discovered that a suspect had connected a USB drive shortly before sensitive files were copied. Even though the device was no longer present, the USBSTOR registry key revealed the device's serial number and connection timestamps, linking the suspect to the activity.
 - Why is the Windows Registry considered a “goldmine” for forensic evidence?
 - How can registry timestamps help reconstruct a timeline of events?
 - What challenges might arise if a suspect attempts to clean or modify the registry?
- In a suspected insider attack, investigators analyzed security logs and found repeated failed login attempts followed by a successful login at an unusual hour. Event ID 4625 entries showed multiple failures, while Event ID 4624 confirmed eventual access. This pattern indicated a password-guessing attempt that ultimately succeeded.
 - What is the difference between System, Application, and Security logs?
 - How can Event IDs help in identifying suspicious behavior?
 - What limitations might arise if logging is disabled or logs are overwritten?

Prefetch Files

- Prefetch files are used by Windows to speed up application startup by caching information about frequently executed programs.
- These files are stored in the C:\Windows\Prefetch directory and have a .pf extension.
- Each prefetch file contains metadata such as the executable name, number of times it has been run, and timestamps of execution. Forensic investigators use prefetch files to determine whether a program was executed on a system, even if the executable itself has been deleted.
- Prefetch analysis can reveal patterns of program usage and help identify malicious activity. For example, if malware was executed, a corresponding prefetch file may still exist, indicating its presence and execution history.
- However, prefetch files may be disabled on some systems or cleared by attackers, so they should be corroborated with other evidence sources.

Prefetch Files

Items: 276

☐	Type	Name	Created (UTC)	Modified (UTC)	Access time (U...
☐	File	SIHCLIENT.EXE-A872A8BF.pf	2021-02-13 7:51:50 F	2021-02-14 2:04:36 AM	2021-02-14 2:04
☐	File	SIHOST.EXE-2C4C53BA.pf	2020-08-01 10:17:19	2020-08-07 5:44:56 PM	2021-02-10 8:03
☐	File	SLUI.EXE-724E99D9.pf	2021-02-13 7:52:27 F	2021-02-14 2:33:33 AM	2021-02-14 2:33
☐	File	SMARTSCREEN.EXE-9B5E4173.pf	2020-08-01 10:02:49	2021-02-14 2:24:10 AM	2021-02-14 2:24
☐	File	SMSS.EXE-E9C28FC6.pf	2020-08-01 6:50:57 F	2021-02-14 2:24:39 AM	2021-02-14 2:24
☐	File	SOFTWARE_REPORTER_TOOL.EXE-0A98C479.pf	2020-08-09 11:06:00	2020-12-10 7:13:14 PM	2021-02-10 7:53
☐	File	SPEECHRUNTIME.EXE-22A4873B.pf	2021-02-13 7:57:19 F	2021-02-14 2:32:42 AM	2021-02-14 2:32
☐	File	SPPEXTCOMOBJ.EXE-BB03B3D6.pf	2021-02-13 7:52:34 F	2021-02-14 2:32:24 AM	2021-02-14 2:32
☐	File	SPPSVC.EXE-B0F8131B.pf	2021-02-13 7:51:18 F	2021-02-14 2:35:02 AM	2021-02-14 2:35
☐	File	SSVAGENT.EXE-BC0480CF.pf	2020-08-07 6:26:32 F	2020-08-07 6:26:32 PM	2021-02-10 8:03
☐	File	SVCHOST.EXE-109E4311.pf	2021-02-13 7:51:18 F	2021-02-14 2:10:05 AM	2021-02-14 2:10
☐	File	SVCHOST.EXE-25616620.pf	2020-08-01 10:17:19	2020-08-07 5:44:56 PM	2021-02-10 8:03

User Activity Artifacts

- User activity artifacts provide insight into how a user interacted with a system. These artifacts include recent files lists, browser history, shortcut (.lnk) files, jump lists, recycle bin contents, and thumbnail caches.
- Together, they allow investigators to reconstruct user behavior, such as which files were opened, which websites were visited, and what actions were performed.
- For example, shortcut files can reveal the original location of a file and the last time it was accessed, even if the file itself has been deleted.
- Jump lists, introduced in newer Windows versions, track frequently accessed files and applications. Browser artifacts can show search queries, downloads, and session timestamps.
- These artifacts are particularly valuable in cases involving data exfiltration, fraud, or insider threats.
- ***On Windows 7 to 11:***
 - ***C:\%USERPROFILE%\AppData\Roaming\Microsoft\Windows\Recent***
 - ***C:\%USERPROFILE%\AppData\Roaming\Microsoft\Office\Recent***

Linux Log Files (/var/log)

- In Linux systems, log files are primarily stored in the /var/log directory and serve as a critical source of forensic evidence. These logs capture system events, authentication attempts, kernel messages, and application activity.
- Key log files include auth.log (authentication events), syslog (general system messages), and kern.log (kernel-related events). The exact file names and locations may vary depending on the Linux distribution.
- Linux logs are typically plain text files, making them easy to parse but also vulnerable to tampering if proper logging protections are not in place.
- Investigators analyze these logs to detect unauthorized access, privilege escalation, and system misuse.
- Tools like grep, awk, and specialized forensic frameworks can be used to filter and analyze large volumes of log data efficiently.

Linux Log Files (/var/log)

```
ubuntu@ip-172-31-11-241:~$  
ubuntu@ip-172-31-11-241:~$ cd /var/log  
ubuntu@ip-172-31-11-241:/var/log$ ls  
alternatives.log      btmp.1      dpkg.log.8.gz      news  
alternatives.log.1    cloud-init.log dpkg.log.9.gz      puppet  
alternatives.log.2.gz ConsoleKit   fontconfig.log     rsyslog-stats  
alternatives.log.3.gz datasync     fsck                syslog  
alternatives.log.4.gz dist-upgrade kern.log           syslog.1  
alternatives.log.5.gz dmesg       kern.log.1         syslog.2.gz  
alternatives.log.6.gz dmesg.0     kern.log.2.gz      syslog.3.gz  
alternatives.log.7.gz dmesg.1.gz  kern.log.3.gz      syslog.4.gz  
alternatives.log.8.gz dmesg.2.gz  kern.log.4.gz      syslog.5.gz  
apache2              dmesg.3.gz  landscape           syslog.6.gz  
appport.log          dmesg.4.gz  lastlog             syslog.7.gz  
appport.log.1        dpkg.log     mail.err            sysstat  
apt                  dpkg.log.1  mail.err.1         tomcat6  
auth.log             dpkg.log.10.gz mail.err.2.gz      udev  
auth.log.1           dpkg.log.2.gz mail.err.3.gz      ufw.log  
auth.log.2.gz        dpkg.log.3.gz mail.log            unattended-upgrades  
auth.log.3.gz        dpkg.log.4.gz mail.log.1         upstart  
auth.log.4.gz        dpkg.log.5.gz mail.log.2.gz      wtmp  
boot.log             dpkg.log.6.gz mail.log.3.gz      wtmp.1  
btmp                 dpkg.log.7.gz mail.log.4.gz  
ubuntu@ip-172-31-11-241:/var/log$
```

Linux Log Files (/var/log)

- Some of the most important Linux system logs include:
- /var/log/syslog and /var/log/messages store all global system activity data, including startup messages. Debian-based systems like Ubuntu store this in /var/log/syslog, while Red Hat-based systems like RHEL or CentOS use /var/log/messages.
- /var/log/auth.log and /var/log/secure store all security-related events such as logins, root user actions, and output from pluggable authentication modules (PAM). Ubuntu and Debian use /var/log/auth.log, while Red Hat and CentOS use /var/log/secure.
- /var/log/kern.log stores kernel events, errors, and warning logs, which are particularly helpful for troubleshooting custom kernels.
- /var/log/cron stores information about scheduled tasks (cron jobs). Use this data to verify your cron jobs are running successfully.

QnA

During a malware investigation, analysts found no trace of the malicious executable on disk. However, a prefetch file named after the malware was discovered, showing it had been executed multiple times. This provided crucial evidence of compromise and helped establish a timeline of infection.

- What type of information can be extracted from a prefetch file?
- How can prefetch files indicate program execution even after deletion?
- Why should prefetch evidence be corroborated with other artifacts?
- What are some common user activity artifacts in Windows systems?
- What artifacts prove a user executed malware?

In a server breach investigation, analysts examined the auth.log file and discovered repeated SSH login attempts from a remote IP address, followed by a successful login. Further analysis revealed that the attacker escalated privileges and installed malicious scripts, all recorded in system logs.

- What types of information can be found in /var/log/auth.log?
- How can Linux logs help detect brute-force attacks?
- What are the risks if log files are not properly secured?

Case Study: Sony Pictures hack

- The Sony Pictures breach of 2014 was one of the most devastating cyberattacks against a major corporation. Attackers infiltrated Sony Pictures Entertainment and gained access to vast amounts of sensitive data, including unreleased films, employee records, internal emails, and executive communications.
- The attack led to widespread operational disruption, reputational damage, and financial losses. The group behind the attack, often referred to as “Guardians of Peace,” deployed destructive malware that wiped systems and rendered them unusable.
- The attackers used sophisticated malware designed not only to infiltrate but also to maintain long-term access within Sony’s network. Persistence mechanisms allowed the attackers to remain undetected for weeks or even months before launching the final destructive phase. The malware created backdoors and likely leveraged scheduled tasks and compromised credentials to ensure continued access even if some components were removed.

Case Study: Sony Pictures hack

- In forensic analysis, persistence was evident through repeated unauthorized access and the presence of malicious binaries across multiple systems. The attackers also used techniques to spread laterally, ensuring that even if one system was cleaned, others remained infected. This demonstrates how persistence is often layered and redundant in advanced attacks.
- Investigators discovered that the attackers had maintained access long enough to map the internal network, escalate privileges, and prepare for data exfiltration and system destruction.
- One of the major challenges in the Sony breach was that attackers deliberately erased or corrupted logs to hinder investigation. Despite this, forensic analysts were able to reconstruct parts of the attack timeline by correlating available system logs, network traffic data, and external monitoring sources. This process revealed how the attackers initially gained entry, moved laterally, and eventually deployed destructive malware.

Case Study: Sony Pictures hack

- Why would attackers maintain persistence for an extended period before acting?
- What indicators might reveal hidden persistence mechanisms in a corporate network?
- How can defenders detect persistence before major damage occurs?
- How can investigators reconstruct events when logs are missing or deleted?
- What types of logs are most critical in large-scale breaches?
- Why is centralized logging important for incident response?
- What factors can lead investigators to suspect an insider threat?
- How can attackers mimic legitimate user behavior?
- What evidence is needed to confirm or rule out insider involvement?

Advanced Windows & Linux Forensics Techniques

- Registry Hive AnalysisRegistry hive analysis involves examining offline copies of Windows registry files such as SYSTEM, SOFTWARE, SAM, SECURITY, and NTUSER.DAT to uncover system and user activity.
- Each hive provides a different perspective: the SYSTEM hive reveals hardware and boot configuration, SOFTWARE shows installed applications, SAM contains user account data, and NTUSER.DAT stores per-user activity such as recent files and preferences.
- From a forensic perspective, investigators extract these hives and analyze key artifacts like USB device history (USBSTOR), auto-start entries (Run keys), and last logged-in users.
- Timestamps embedded in registry keys (LastWrite times) are critical for building timelines. Even if a system is wiped or partially cleaned, these hives often retain historical evidence that can reveal prior activity.

Advanced Windows & Linux Forensics Techniques

- **Event ID Analysis**
- Event ID analysis focuses on interpreting specific Windows Event Log codes to identify system and user actions. Each Event ID corresponds to a particular activity—for example:
 - 4624 → Successful logon
 - 4625 → Failed logon
 - 4688 → Process creation
 - 1102 → Log cleared
- By analyzing these IDs, investigators can quickly detect suspicious patterns such as brute-force attacks, privilege escalation, or log tampering. Event logs also include timestamps, user accounts, and source systems, making them essential for tracing attacker behavior.

Advanced Windows & Linux Forensics Techniques

- **Timeline Creation**

- Timeline creation is the process of merging multiple data sources into a chronological sequence of events.
- Investigators collect timestamps from registry hives, event logs, file systems, prefetch files, and network logs, then normalize them into a unified timeline.
- This allows analysts to see cause-and-effect relationships, such as login → file access → data transfer.
- A well-constructed timeline helps answer key questions: when the attack started, how it progressed, and what actions were taken.
- Tools like Plaso or Autopsy can automate timeline generation, but analysts must still interpret and validate the results.

Advanced Windows & Linux Forensics Techniques

- **User Account Investigation**
- User account investigation focuses on analyzing how accounts are created, used, and potentially abused. Investigators examine login activity, privilege levels, group memberships, and account changes.
- Key artifacts include Security Event Logs, the SAM hive, and Linux authentication logs.
- Suspicious indicators include:
 - Logins at unusual times
 - Use of dormant or disabled accounts
 - Privilege escalation (e.g., standard user becoming admin)
 - Multiple logins from different locations
- This analysis helps determine whether an account was legitimately used, compromised, or maliciously created.

Advanced Windows & Linux Forensics Techniques

- **Artifact Correlation Flow (Demonstration)**
- Artifact correlation is the process of linking multiple forensic data sources to form a complete narrative of an incident. No single artifact tells the full story—only by combining them can investigators reach reliable conclusions.
 - Step 1: Security logs show a successful login (Event ID 4624) at an unusual time.
 - Step 2: Registry hive (NTUSER.DAT) shows recent file access entries for sensitive documents.
 - Step 3: Prefetch files indicate that a file archiving tool (e.g., WinRAR) was executed shortly after login.
 - Step 4: Registry (USBSTOR) reveals a USB device connected minutes later.
 - Step 5: System and network logs show data transfer activity or device mounting.
 - Step 6: All timestamps are merged: 02:00 → Login; 02:05 → File access; 02:07 → Archive tool execution; 02:10 → USB connection;

END